Tuesday
22/04/25

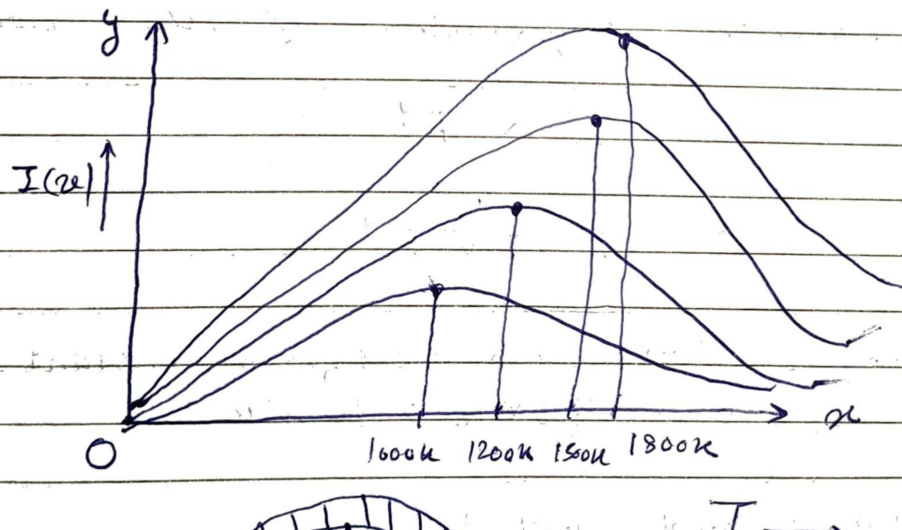
Physics

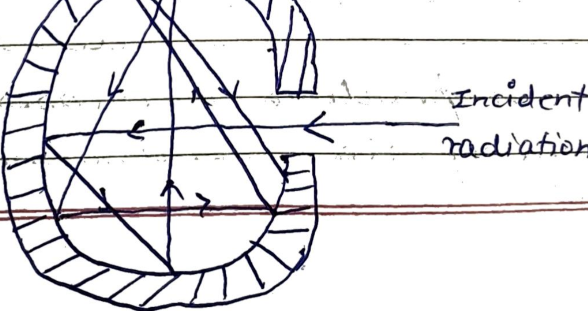
Quantum Mechanics

• **Black Body Radiation :-**

A body that absorbs the entire radiation incident on it is called a perfect black body. When a black body is heated, it emits radiation at all frequency, and thus it is a good emmitter as well as good absorber. The radiation emitted by perfect black body is called black body radiation.

The experimental measure of intensity $I(\nu)$ with different frequency is shown in figure for different value of temperature. This plot shows that





↳ The distribution of frequency

↳ With the increase of temperature, the total amount of radiation increases.

↳ The position of the maxima peak shift towards higher frequency with increase in equilibrium temperature.

• **Laws of Black Body Radiation :-**

(1) Stefan - Boltzman Law :-

It states that the total radiation emitted from a black body at a temperature T is proportional to the fourth power of the absolute temperature of the body.

$$E \propto T^4$$

$$\Rightarrow E = \sigma T^4$$

Where $\sigma \rightarrow$ Stefan's - Boltzman constant

$\sigma = 5.6 \times 10^{-8}\ W m^{-2} K^{-4}$

(2) Wein's Displacement Law :-

Wein's states that as the temperature increases the wavelength corresponding to maximum energy (λ_m) decreases.

$$\lambda_m \propto \frac{1}{T}$$

$$\lambda_m T = \text{constant}$$

$$\boxed{\lambda_m T = b}$$

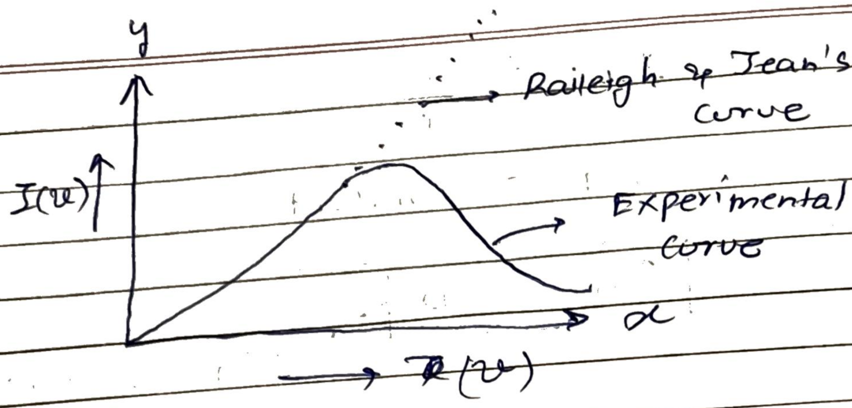
Wein's Constant.

<3> Rayleigh Jean's law & Ultraviolet Catastroph :-

↳ According to them, electromagnetic radiation inside the cavity of the black body at an equilibrium temperature T forms the standing wave and the possible modes in the cavity is large if the wavelength is small. So the increase in the number of modes is proportional to $\frac{1}{\lambda^2}$ and v^2 and also each of the standing wave must be assigned an average kinetic energy KT where K is the Boltzman's constant. This leads to the following Railey-Jean's Law,

$$I(v)\,dv = \frac{8\pi v^2 KT\,dv}{c^3}$$

↳ This relation shows that $I(v)$ is proportional to the square of v. The corresponding plot is shown in figure.



It is clear that the experimental data does'nt agree with the theory. The agreement is good only for smaller value of ν and the disagreement at higher frequency i.e., in the U.V region is called ultraviolet catastroph

Date
Page

Monday
/05/25

Physics
Quantum Mechanics

- Plank's Theory of Radiation :-
- Plank's Quantum Hypothesis:-
- **Avg. Energy of the Plank's oscillator:-**

↳ If n be the total number of oscillator & E is the total energy of these oscillator Then the avg energy is given by the relation,

$$\bar{E} = \frac{E}{N}$$

Let $N_0, N_1, N_2, \ldots, N_n$ are the number of oscillator having energy value $0, h\nu, 2h\nu, \ldots n h\nu$. Then by Maxwell's distribution formula,

$$N_n = N_0 e^{-nh\nu/kT}$$

If $N = N_0 + N_1 + N_2 + \ldots + N_n$

$\Rightarrow N = N_0 + N_0 e^{-h\nu/kT} + N_0 e^{-2h\nu/kT} + \ldots N_0 e^{-nh\nu/kT}$

$\Rightarrow N = N_0 (1 + e^{-h\nu/kT} + e^{-2h\nu/kT} + \ldots + e^{-nh\nu/kT})$

$\Rightarrow N = \dfrac{N_0}{1 - e^{-h\nu/kT}}$ [For G.P.

Total energy $(E) = (N_0 \times 0) + (N_1 \times h\nu) + (N_2 \times 2h\nu) + \ldots (N_n \times nh\nu)$

$\Rightarrow E = 0 + N_0 e^{-h\nu/kT} h\nu + N_0 e^{-2h\nu/kT} \cdot 2h\nu + \ldots$
$+ N_0 e^{-nh\nu/kT} \times nh\nu.$

$\Rightarrow E = N_0 \left(e^{-h\nu/kT} h\nu + 2e^{-2h\nu/kT} h\nu + 3e^{-3h\nu/kT} h\nu + \dots n e^{-nh\nu/kT} h\nu \right)$

$\Rightarrow$ ~~$E = N_0$~~

$\Rightarrow$ ~~$E \times$~~ $e^{-h\nu/kT} = N_0 \left(e^{-2h\nu/kT} + 2e^{-3h\nu/kT} + 3e^{-4h\nu/kT} + \dots (n-1)e^{-nh\nu/kT} + ne^{-(n+1)h\nu/kT} \right)$

Let,

$$e^{-h\nu/kT} = x$$

$$\Rightarrow \therefore E = N_0 e^{-h\nu/kT} h\nu \left(1 + 2x + 3x^2 + 4x^3 + \dots nx^{n-1} \right)$$

Now,

$$\bar{E} = \frac{E}{N} = \frac{\cancel{N_0} h\nu e^{-h\nu/kT} (1+2x+3x^2+4x^3+\dots nx^{n-1})}{\cancel{N_0} (1+x+x^2+x^3+\dots x^n)}$$

$$\therefore \bar{E} = \frac{h\nu x (1-x)^{-2}}{(1-x)^{-1}}$$

$$\Rightarrow \bar{E} = \frac{h\nu x}{1-x}$$

- <u>Plank's</u> <u>Radiation</u> <u>formula</u>:-

The energy density of radiation $u(\nu)$ in the frequency range ν and $\nu + d\nu$ depending upon the average energy of osillation and is given by

$$u(\nu) = \frac{8\pi\nu^2}{c^3} \bar{E}$$

$$\Rightarrow u(\nu)\, d\nu = \frac{8\pi\nu^2}{c^3} \bar{E}\, d\nu$$

$$\Rightarrow u(\nu)\,d\nu = \frac{8\pi\nu^2}{c^3} \times \frac{h\nu x}{1-x}\,d\nu$$

$$\Rightarrow u(\nu)\,d\nu = \frac{8\pi\nu^3 h}{c^3} \times \frac{1}{(e^{h\nu/kT}-1)}\,d\nu$$

The above relation is known as plank radiation formula in terms of frequency.

In terms of wavelength,

$$\Rightarrow u(\lambda) \times \frac{-\cancel{c}}{\lambda^2}\,d\lambda = \frac{8\pi \cancel{c^3}}{\lambda^3 \times \cancel{c^3}} \times h \times \frac{1}{(e^{+hc/\lambda kT}-1)} \times \frac{-\cancel{c}}{\cancel{\lambda^2}}\,d\lambda$$

$$\Rightarrow u(\lambda)\,d\lambda = \frac{8\pi h}{\lambda^3} \times \frac{1}{(e^{+hc/\lambda kT}-1)}\,d\lambda$$

- <u>Wein's Law</u> :-

↳ It is for high frequency or lower wavelength

$$(e^{h\nu/kT}-1) \simeq e^{h\nu/kT}$$

$$\Rightarrow \therefore u(\nu)\,d\nu = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{h\nu/kT}}\,d\nu$$

- <u>Rayleigh's Jeans Law</u> :-

↳ This is for lower wavelength or higher wavelength.

$\therefore \; e^{h\nu/kT} = 1 + \frac{h\nu}{kT}$

$\Rightarrow$ So, $u(\nu)\,d\nu = \frac{8\pi h\nu^3}{c^3} \times \frac{1}{\left(\cancel{1} + \frac{h\nu}{kT} \cancel{-1}\right)} \, d\nu$

$\Rightarrow \; u(\nu)\,d\nu = \frac{8\pi h\nu^2 kT}{c^3 \times h} d\nu = \frac{8\pi\nu^2 kT}{c^3} d\nu$

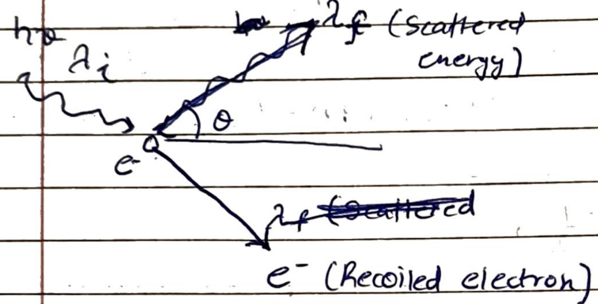
* Rest mass of photon is zero, they are neutral, they travel with the speed of light and possess momentum because of the energy.

Tuesday
29/04/25

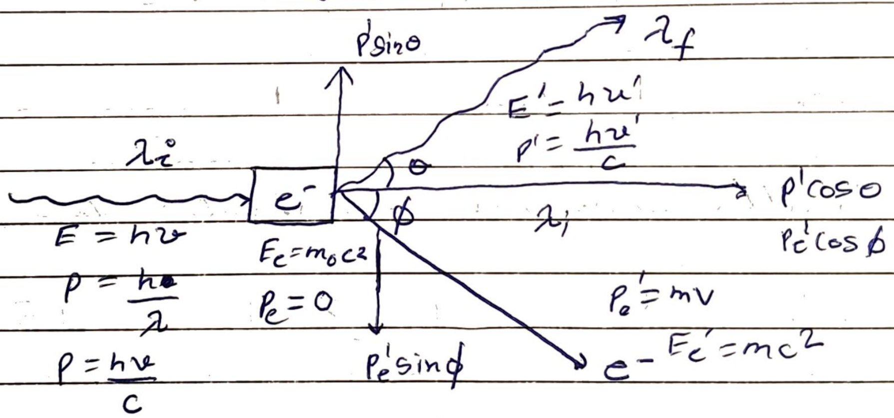
Physics

Quantum Mechanics

• **Comton's Effect :-** [or Comton's Scattering]
↳ wavelength depends on the scattering angle.



$$\Rightarrow \quad \Delta\lambda = \lambda_f - \lambda_i = \frac{h}{m_0 c}(1-\cos\theta)$$



Here, $m = \dfrac{m_0}{\sqrt{1 - V^2/c^2}}$ & $\nu > \nu'$, $\lambda_i < \lambda_f$

By the conservation of energy,

$$E_T \text{ (Before)} = E_T \text{ (After)}$$

$$\therefore \quad h\nu + m_0 c^2 = h\nu' + mc^2$$

$$\Rightarrow \quad h\nu - h\nu' = (m - m_0)c^2$$

$$\Rightarrow h(\nu - \nu') = (m - m_0) c^2$$

$$\Rightarrow h\cancel{c}\left(\frac{1}{\lambda} - \frac{1}{\lambda'}\right) = \left(\frac{m_0}{\sqrt{1-\frac{v^2}{c^2}}} - m_0\right) c^{\cancel{2}}$$

$$\Rightarrow h\left(\frac{1}{\lambda} - \frac{1}{\lambda'}\right) = m_0 c\left(\frac{1}{\sqrt{1-v^2/c^2}} - 1\right) \quad \text{--- (1)}$$

According to the Law of conservation of momentum.

In the direction of incident photon,

$$p + p_e = p' \cos\theta + p_e' \cos\phi$$

$$\Rightarrow \frac{h}{\lambda} + 0 = \frac{h}{\lambda'}\cos\theta + mv\cos\phi \quad \text{--- (2)}$$

In the direction perpendicular to the incident photon,

$$\Rightarrow 0 = p'\sin\theta + p_e'\sin\phi\,(-1)$$

$$\Rightarrow \frac{h}{\lambda'}\sin\theta = mv\sin\phi \quad \text{--- (3)}$$

On squaring & adding (2) & (3),

$$\Rightarrow m^2v^2 = \left(\frac{h}{\lambda} - \frac{h}{\lambda'}\cos\theta\right)^2 + \left(\frac{h}{\lambda'}\sin\theta\right)^2$$

$$\Rightarrow m^2v^2 = \frac{h^2}{\lambda^2} + \frac{h^2}{\lambda'^2}\cos^2\theta - \frac{2h^2\cos\theta}{\lambda\lambda'} + \frac{h^2}{\lambda'^2}\sin^2\theta$$

$$\Rightarrow m^2v^2 = \frac{h^2}{\lambda^2} + \frac{h^2}{\lambda'^2} - \frac{2h^2\cos\theta}{\lambda\lambda'}$$

$$\Rightarrow m^2v^2 = h^2\left[\left(\frac{1}{\lambda^2} + \frac{1}{\lambda'^2}\right) - \frac{2\cos\theta}{\lambda\lambda'}\right]$$

$$\Rightarrow \frac{m_0^2}{1-\frac{v^2}{c^2}} \times v^2 = h^2\left[\left(\frac{1}{\lambda^2}+\frac{1}{\lambda'^2}\right) - \frac{2\cos\theta}{\lambda\lambda'}\right]$$

$$\Rightarrow \frac{m_0^2 c^2 v^2}{c^2 - v^2} = h^2\left[\left(\frac{1}{\lambda^2}+\frac{1}{\lambda'^2}\right) - \frac{2\cos\theta}{\lambda\lambda'}\right] \quad —④$$

From eq^n-①, we have,

$$\frac{h}{\lambda} - \frac{h}{\lambda'} + m_0 c = \frac{m_0 c}{\sqrt{1-v^2/c^2}} \quad *$$

Squaring both sides,

$$\Rightarrow \cancel{\frac{h^2}{\lambda^2}} + \left(\frac{h}{\lambda} - \frac{h}{\lambda'}\right)^2 + m_0^2 c^2 + 2m_0 ch\left(\frac{1}{\lambda} - \frac{1}{\lambda'}\right) = \frac{m_0^2 c^2}{1 - v^2/c^2}$$

$$\Rightarrow \frac{h^2}{\lambda^2} + \frac{h^2}{\lambda'^2} - \frac{2h^2}{\lambda\lambda'} + m_0^2 c^2 + 2m_0 ch\left(\frac{1}{\lambda} - \frac{1}{\lambda'}\right) = \frac{m_0^2 c^2}{1 - v^2/c^2}$$

$$\Rightarrow \frac{h^2}{\lambda^2} + \frac{h^2}{\lambda'^2} - \frac{2h^2}{\lambda\lambda'} + 2m_0 ch\left(\frac{1}{\lambda} - \frac{1}{\lambda'}\right) = m_0^2 c^2\left[\frac{c^2}{c^2 - v^2} - 1\right]$$

$$= m_0^2 c^2\left[\frac{v^2}{c^2 - v^2}\right]$$

$$\Rightarrow \frac{h^2}{\lambda^2} + \frac{h^2}{\lambda'^2} - \frac{2h^2}{\lambda\lambda'} + 2m_0 ch\left(\frac{1}{\lambda} - \frac{1}{\lambda'}\right) = \frac{m_0^2 c^2 v^2}{c^2 - v^2} \quad —⑤$$

Putting this value in eq^n-④,

$$\Rightarrow \cancel{h^2\left(\frac{1}{\lambda^2}+\frac{1}{\lambda'^2}\right)} - \frac{2h^2}{\lambda\lambda'} + 2m_0 ch\left(\frac{1}{\lambda} - \frac{1}{\lambda'}\right) = h^2\left[\cancel{\left(\frac{1}{\lambda^2}+\frac{1}{\lambda'^2}\right)} - \frac{2\cos\theta}{\lambda\lambda'}\right]$$

$$\Rightarrow \frac{-2h^2}{\lambda\lambda'} + 2m_0 hc\left(\frac{1}{\lambda} - \frac{1}{\lambda'}\right) = -\frac{2h^2\cos\theta}{\lambda\lambda'}$$

Date
Page

$$\Rightarrow \quad 2m_o \cancel{h} c\left(\frac{1}{\lambda} - \frac{1}{\lambda'}\right) = \frac{2h^{\cancel{2}}}{\lambda\lambda'}(1-\cos\theta)$$

$$\Rightarrow \quad 2m_o c\left(\frac{\lambda' - \lambda}{\cancel{\lambda\lambda'}}\right) = \frac{2h}{\cancel{\lambda\lambda'}}(1-\cos\theta)$$

$$\Rightarrow \quad \lambda' - \lambda = \frac{2h}{2m_o c}(1-\cos\theta)$$

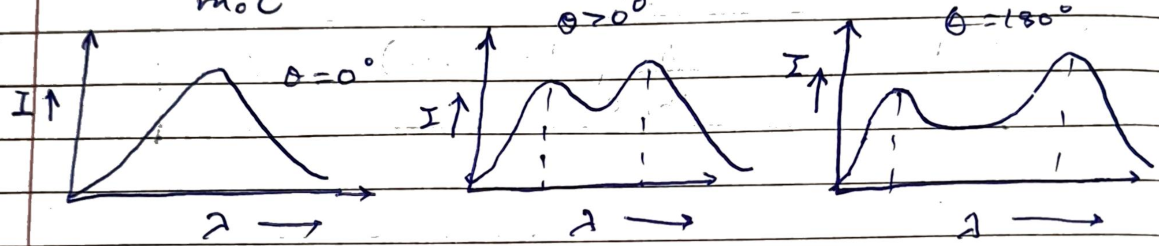
$$\Rightarrow \quad \boxed{\lambda' - \lambda = \frac{h}{m_o c}(1-\cos\theta) = \Delta\lambda}$$

Thursday
1/05/25

Physics
Quantum Mechanics

- **Why Crompton effect is not visible for visible light?**

$$\Delta\lambda = \frac{h}{m_0 c}(1-\cos\theta) = 0.0242\,(1-\cos\theta)\ \text{Å}$$



For visible light, $\lambda \in (4000\,\text{Å},\ 7000\,\text{Å})$

for $\lambda = 4000\,\text{Å}$,

$$\frac{\Delta\lambda}{\lambda}\times 100 = \frac{0.04}{4000}\times 100 = 0.001$$

for $\lambda = 7000\,\text{Å}$,

$$\frac{\Delta\lambda}{\lambda}\times 100 = \frac{0.04}{7000}\times 100 =$$

- Q. An X-ray photon is found to have doubled its wavelength on being scattered by 90°. Find the energy and wavelength of incident photon.

Soln →

$$\lambda = \frac{h}{m_0 c}(1-\cos\theta) = 0.0242\times(1-0)\ \text{Å}$$

$$\Rightarrow \lambda = 0.0242\ \text{Å}$$

$$E = \frac{hc}{\lambda} = \frac{6.63\times10^{-34}\times 3\times10^{8}}{0.0242\times10^{-10}} = 821.9\times10^{-16}\ \text{J}$$

$$= \frac{821.9\times10^{-16}}{1.6\times10^{-19}} = 513.7\ \text{MeV}$$

Date
Page

- ~~Debrogl~~

doubt Q. X-ray of wavelength 0.2 Å are scattered from a target, calculate the wavelength of X-ray scattered through 45° and also find the maximum kinetic energy of recoiled electron ?

Sol^n → $\lambda_f - 0.2 = 0.0242\left(1 - \frac{1}{\sqrt{2}}\right)$

$\lambda_f = 0.1207$ Å

$E_e = hc\left(\frac{1}{\lambda_i} - \frac{1}{\lambda_f}\right) = 6.63 \times 10^{-34} \times 3 \times 10^{8} \times \left(\frac{10^{10}}{0.2} - \frac{10^{10}}{0.207}\right)$

$\Rightarrow E_e = \frac{19.89 \times 10^{-16} \times 0.169}{1.6 \times 10^{-19}} = 2.1 \times 10^{3}\ eV$

- **Debroglie Hypothesis :-**

↳ Moving material particle is associated with wave ~~associated~~ whose wavelength is given by ;

Debroglie wavelength ← $\boxed{\lambda = \frac{h}{p}}$ → Momentum of particle

$KE = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{\frac{2KE}{m}}$

$P = mv = \sqrt{2mKE}$

$KE = e \times V$ [for charged particle is e^-]

$\Rightarrow \lambda = \frac{h}{\sqrt{2meV}}$

$$\Rightarrow \lambda = \frac{12.27}{\sqrt{V}} \text{ Å}$$

For charged particle q,

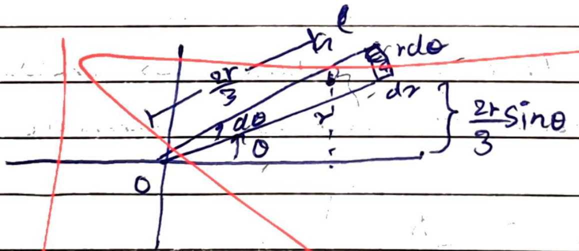
$$\Rightarrow \lambda = \frac{h}{\sqrt{2mqV}}$$

↳ If a material particle is in thermal equilibrium at an absolute temperature T,

$$KE = \frac{3}{2} kT$$

↳ If a particle is moving with relative speed

$$mv = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}} \times c$$



$$dA = \frac{1}{2} r^2 d\theta$$

$$d = 2\pi y_{centroid}$$

$$d = 2\pi \left(\frac{2r \sin\theta}{3} \right)$$

$$V = \int d \cdot dA$$

$$V = \int_{\theta_1}^{\theta_2} 2\pi \left(\frac{2r \sin\theta}{3} \right) \times \frac{1}{2} r^2 d\theta$$

$$V = \frac{2\pi}{3} \int_{\theta_1} r^3 \sin\theta \, d\theta$$

Physics

↳ $\frac{h}{m_o c}$ → Crompton wavelength.

↳ if $\phi = 0°$ ⇒ $\lambda' - \lambda = 0$

↳ if $\phi = 180°$ ⇒ $\lambda' - \lambda = \frac{2h}{m_o c}$ (max possible value)

- **Hiesenberg's Uncertainty Principle :-**

↳ it is only valid on microscopic particles.

↳ $\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$ or $\Delta \theta \, \Delta J \geq \frac{h}{4\pi}$

J → ang' momentum.

↳ $\Delta E \cdot \Delta t \geq \frac{h}{4\pi}$

Q. Δx → Given, Δp → find or vice versa.

Q. prove by HUP that e^- can't be in nucleus.

Q. Δx → Given, Δv → find or vice versa

Q.

Physics

• **De-broglie wavelength :-**

$$\lambda = \frac{h}{p}$$

or $KE = \frac{1}{2}mv^2$ (Classical)

$\Rightarrow KE = \frac{p^2}{2m}$ (Quantum)

So, $\lambda = \frac{h}{\sqrt{2mKE}}$

Also, $eV = E$

So, $\lambda = \frac{h}{\sqrt{2meV}}$

$\lambda = \frac{1.234 \times 10^{-9}}{\sqrt{V}}$ m

or $\lambda = \frac{1.234}{\sqrt{V}}$ nm (for e^-)

↳ Also, $KE = \frac{3}{2}kT$

So, $\lambda = \frac{h}{\sqrt{2m\frac{3kT}{2}}} = \frac{h}{\sqrt{3mkT}}$

• Wave function;

↳ Classical → $y = A\sin\omega t$

$$\hbar = \frac{h}{2\pi} = 1.1 \times 10^{-34} \text{ J}$$

↳ Quantum, $\psi = Ae^{-\frac{i}{\hbar}\underbrace{(Et-px)}_{\text{phase}}}$

imp properties of wave function :-

1) it must be continuous.
2) ψ'_x must also be continuous $\left(\psi' = \frac{\partial \psi}{\partial x}\right)$

3) The wave function must be single valued.

4) wave function must be normalised.

i.e., $\int_{-\infty}^{\infty} |\psi|^2 d\tau = 1$

$\underbrace{|\psi|^2}_{\text{probability of finding the particle.}} = \psi^* \cdot \psi$. ($\psi^* \to i$ is replaced with $-i$)

\# Q. What are wave functions & their properties.

- Operators :-

1) Momentum operator (p)
2) Energy operator (E).

↳ $\psi = Ae^{-\frac{i}{\hbar}(Et-px)}$ (in one-dimension)

So, $\frac{\partial \psi}{\partial x} = \frac{ip}{\hbar} Ae^{-\frac{i}{\hbar}(Et-px)}$

$\Rightarrow \frac{\partial \psi}{\partial x} = \frac{ip}{\hbar}\psi$.

Date______
Page______

or $$\boxed{\begin{aligned} &\frac{\hbar}{i}\frac{\partial\psi}{\partial x} = p\psi \\ \text{or} \quad &-i\hbar\frac{\partial\psi}{\partial x} = p\psi. \end{aligned}}$$

So, $\boxed{p = -i\hbar\frac{\partial}{\partial x}}$ → momentum operator.

for 3-dimensions,

$\boxed{p = -i\hbar\nabla}$ where $\nabla = \frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}$

nebla

4 Now, $\frac{\partial\psi}{\partial t} = -\frac{i}{\hbar}EAe^{-\frac{i}{\hbar}(Et-px)}$

or $$\boxed{\begin{aligned} &-\frac{\hbar}{i}\frac{\partial\psi}{\partial t} = E\psi \\ \text{or} \quad &i\hbar\frac{\partial\psi}{\partial t} = E\psi \end{aligned}}$$

So, $\boxed{E = i\hbar\frac{\partial}{\partial t}}$ → Energy operator.

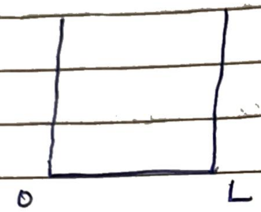
for 3-dimensions,

$\boxed{E = i\hbar\nabla}$ where, $\nabla = \frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}$

Tuesday
6/5/25.

Physics
Quantum Mechanics

- **Wave function (ψ) :-**



$$\int_0^L \psi^* \psi = 1$$

- **Schrödinger Wave Equation :-**

It is the eq^n which represents the wave function (ψ) of matter waves in different physical condition. This eq^n tells the behaviour of wave function associated with the atomic particle. This eq^n can be established just like the eq^n of an ordinary plane progressive wave.

- **Time independent Schrödinger Equation :-**

A general wave equation in differential form is given by $\frac{\partial^2 \psi}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}$ for one dimension

and $\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}$

or $$\nabla^2 \psi = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2} \quad \text{—— (2)}$$

The solution of eq^n-(2) gives ψ as a periodic displacement in terms of ~~sine~~ time, i.e.,

$$\psi(x, y, z, t) = \psi_0 (x, y, z, t)\, e^{-i\omega t}$$
$$= \psi_0 (r, t)\, e^{-i\omega t}$$

Date :
P. No. :

where, ψ_0 is the amplitude of the particle wave at the point (x, y, z) which is independent of time t.

$$\frac{\partial \psi}{\partial t} = \psi_0 e^{-i\omega t} \times (-i\omega) = -i\omega \psi_0 e^{-i\omega t}$$

$$\frac{\partial^2 \psi}{dt^2} = -\omega^2 \psi_0 e^{-i\omega t} = -\omega^2 \psi$$

So,

$$\nabla^2 \psi = \frac{1}{v^2} \times -\omega^2 \psi$$

$$\Rightarrow \nabla^2 \psi = \frac{-\omega^2}{v^2} \psi$$

or $\nabla^2 \psi + \frac{\omega^2}{v^2} \psi = 0$

Now, $\omega = 2\pi f = \frac{2\pi v}{\lambda} \Rightarrow \frac{\omega}{v} = \frac{2\pi}{\lambda}$

$$\Rightarrow \nabla^2 \psi + \frac{4\pi^2 \psi}{\lambda^2} = 0$$

According to De-Broglie, $\lambda = \frac{h}{p}$

$$\Rightarrow \therefore \nabla^2 \psi + \frac{4\pi^2 \psi \times p^2}{h^2} = 0$$

or $\nabla^2 \psi + \frac{4\pi^2 m^2 v^2}{h^2} \psi = 0$ ——— (3)

If E and P are respectively the total energy and potential energy of the particle then its kinetic energy is given by,

$$\frac{1}{2} mv^2 = E - P$$

$$\Rightarrow mv^2 = 2(E - P)$$

$$\Rightarrow m^2 v^2 = 2m(E - P)$$

Put this in eq^n - ③,

$\Rightarrow \nabla^2\psi + \frac{8\pi^2 m}{h^2}(E-V)\psi = 0$ [where V is the potential energy]

$\rightarrow$ or $\boxed{\nabla^2\psi + \frac{2m}{\hbar^2}(E-V)\psi = 0}$

For a free particle, $V = 0$

$\rightarrow \nabla^2\psi + \frac{2mE}{\hbar^2}\psi = 0$.

- Time-dependent Schrodinger Equation :-

We have, $\cancel{\nabla^2\psi = \psi_0}$

$\nabla^2\psi + \frac{2m}{\hbar}(E-V)\psi = 0$ &

$\psi = \psi_0(r)\, e^{-i\omega t}$

$\Rightarrow \frac{d\psi}{dt} = -i\omega\, \psi_0(r)\, e^{-i\omega t} = -i\omega\psi$

Now, $E = h\nu \Rightarrow \nu = \frac{E}{h}$

& $\omega = 2\pi\nu = \frac{2\pi E}{h}$

$\Rightarrow \omega = \frac{E}{\hbar}$

So, $\frac{\partial\psi}{\partial t} = \frac{-iE}{\hbar}\psi$

or $\frac{\partial\psi}{\partial t} = \frac{E}{i\hbar}\psi$

Energy operator

$\Rightarrow i\hbar\frac{\partial\psi}{\partial t} = E\psi$ where $E = i\hbar\frac{\partial}{\partial t}$

momentum operator

& $p = -i\hbar\frac{\partial}{\partial x}$

Date :
P. No. :

Putting this value above,

$$\Rightarrow \nabla^2\psi + \frac{2m}{\hbar}\left(\frac{i\hbar}{\psi}\frac{\partial\psi}{\partial t} - V\right)\psi = 0$$

$$\Rightarrow \nabla^2\psi + 2mi\frac{\partial\psi}{\partial t} - \frac{2m}{\hbar}\psi V = 0$$

$$\Rightarrow \boxed{\frac{-\hbar}{2m}\Delta^2\psi + V\psi = i\hbar\frac{\partial\psi}{\partial t}}$$

Physics

- ~~Shro~~ Schrodinger's wave equation
  - Time dependent
  - Time independent.

↳ T.E. = KE + PE.

↳ in quantum, $H = KE + V$ → PE.
Hamiltonian

So, $H = TE$.

$\Rightarrow TE = \frac{p^2}{2m} + V$

Let $TE = E$

$\Rightarrow E = \frac{p^2}{2m} + V$

Now,

$E\psi = \frac{p^2\psi}{2m} + V\psi$ (operating wave function)

$E\psi = \frac{p \cdot p\psi}{2m} + V\psi$

$\Rightarrow$ ~~$E\psi = p(-i\hbar$~~ $p\psi = -i\hbar\frac{\partial\psi}{\partial x}$

$p^2\psi = + i^2\hbar^2\frac{\partial^2\psi}{\partial x^2}$

So,

$i\hbar\frac{\partial\psi}{\partial t} = \frac{-\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V\psi$.

$$E\psi = \frac{-\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V\psi.$$

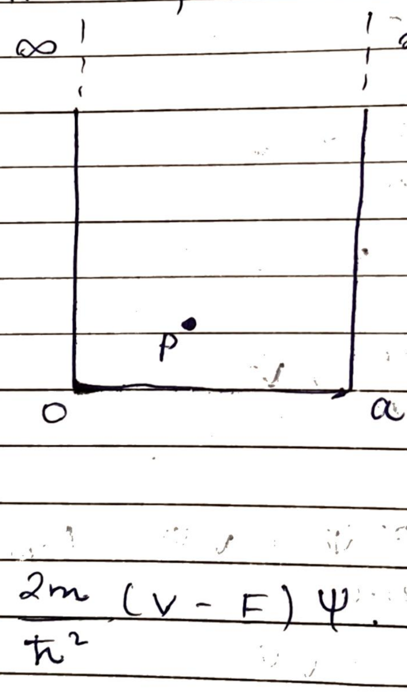
$$\Rightarrow \frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} = (V-E)\psi$$

$$\Rightarrow \boxed{\frac{\partial^2\psi}{\partial x^2} = \frac{2m}{\hbar^2}(V-E)\psi}$$

(Schrodinger's time independent equation.

- Applications of Schrodinger's wave Equation (Time independent) :-

(i) Particle in ∞ potential ~~wortot~~ well (Box)



$$V = \begin{cases} 0 ; & 0 < x < a \\ \infty ; & \cancel{x > a} \\ & x \le 0, x \ge a \end{cases}$$

$\psi = ?$, $E = ?$

We have

$$\frac{\partial^2\psi}{\partial x^2} = \frac{2m}{\hbar^2}(V-E)\psi.$$

$$\Rightarrow \frac{\partial^2\psi}{\partial x^2} - \frac{2m}{\hbar^2}(V-E)\psi = 0$$

$$\Rightarrow \frac{\partial^2\psi}{\partial x^2} + \frac{2mE\psi}{\hbar^2} = 0 \qquad (V=0 \text{ for free particle})$$

Let $K^2 = \frac{2mE}{\hbar^2}$

$$\frac{d}{dt}\left(\frac{d\psi}{dt}\right)$$

$$\frac{d\psi}{d\psi} + \frac{d\psi}{d\psi}$$

$\Rightarrow$ $$\frac{\partial^2 \psi}{\partial x^2} + k^2\psi = 0.$$

Standard solⁿ

$$\psi = A\sin kx + B\cos kx.$$

By using boundry conditions, at $x = 0$

$\psi = 0$

$\Rightarrow$ $0 = A\sin k(0) + B\cos k(0)$

$\Rightarrow$ $0 = 0 + B$

$\Rightarrow$ $\boxed{B = 0}$

At $x = a$,

$\psi = 0$

$\Rightarrow$ $0 = A\sin ka$

$\Rightarrow$ $A = 0$ or $\sin ka = 0$

(not possible) $\Rightarrow ka = n\pi$

$\Rightarrow \boxed{k = \frac{n\pi}{a}}$

So, $$\left(\frac{n\pi}{a}\right)^2 = \frac{2mE}{\hbar^2}$$

$\Rightarrow$ $$\boxed{E = \frac{n^2\pi^2\hbar^2}{2ma^2}}$$

$$\boxed{E = \frac{n^2h^2}{8ma^2}}$$

Date
Page

So, we have $\psi = A\sin\left(\frac{n\pi x}{a}\right)$.

By normalising.

So, $\int_0^a |\psi|^2 dx = 1$

$\Rightarrow A^2 \int_0^a \sin^2\left(\frac{n\pi x}{a}\right) dx = 1$

$\frac{A^2}{2} \int_0^a \left(1 - \cos\left(\frac{2n\pi x}{a}\right)\right) dx = 1$

$\Rightarrow \frac{A^2}{2}[a-0] - \frac{A^2}{2} \times \frac{a}{2n\pi} \left[\sin\left(\frac{2n\pi x}{a}\right)\right]_0^a = 1$

$\Rightarrow \frac{A^2 a}{2} - \frac{A^2 a}{4n\pi}\left[\sin(2n\pi) - \sin 0\right] = 1$

$\frac{A^2 a}{2} = 1$

$A^2 = \frac{2}{a}$

$\boxed{A = \sqrt{\frac{2}{a}}}$

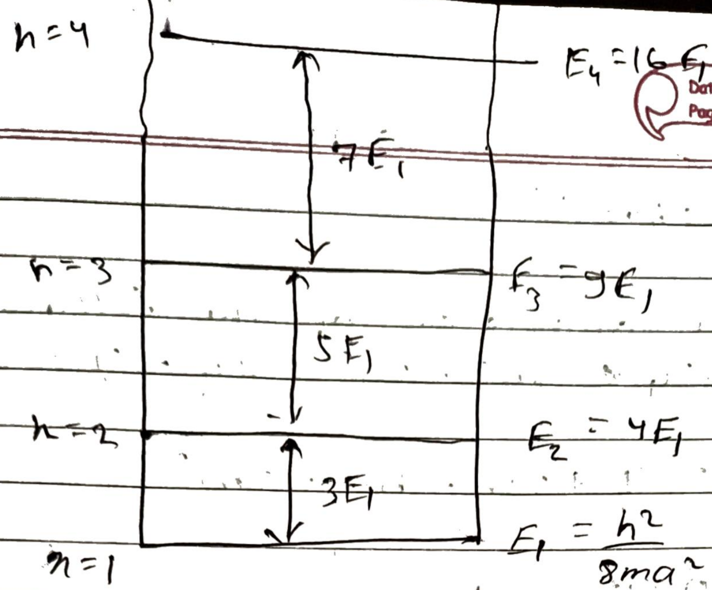
01/05/25

↳ $\psi(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)$!

↳ $E_n = \frac{n^2 h^2}{8ma^2}$, $n = 1, 2, 3, 4,$

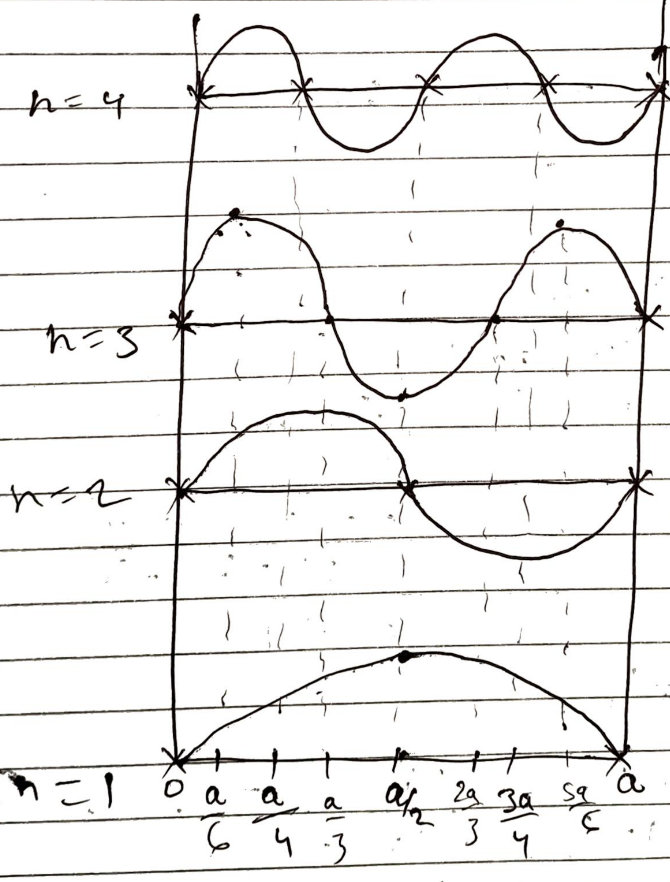
$n \neq 0$

Date
Page



$$\psi_1 = \sqrt{\frac{2}{a}} \sin\left(\frac{\pi x}{a}\right), \quad \psi_2 = \sqrt{\frac{2}{a}} \sin\left(\frac{2\pi x}{a}\right)$$

$$\psi_3 = \sqrt{\frac{2}{a}} \sin\left(\frac{3\pi x}{a}\right), \quad \psi_4 = \sqrt{\frac{2}{a}} \sin\left(\frac{4\pi x}{a}\right)$$



$\frac{4\pi x}{a} = \pi$

$x = \frac{a}{4}$

$\frac{4\pi x}{a} = \frac{\pi}{2}$

$x = \frac{a}{8}$

$\frac{3\pi x}{a} = \pi$

$x = \frac{a}{3}$

$\frac{3\pi x}{a} = \frac{\pi}{2}$

$x = \frac{a}{6}$

$\frac{3\pi x}{a} = \frac{3\pi}{2}$

$\boxed{x = \frac{a}{2}}$

$\frac{3\pi x}{a} = 2\pi$

$x = \frac{2a}{3}$

$\frac{4\pi x}{a} = \frac{5\pi}{2}$

$\frac{5\pi}{2}$

0.6

0.7

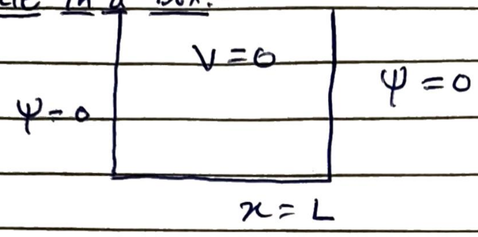
Date :
P. No. :

Thursday
8/05/25

Physics

Quantum Mechanics

• Particle in a Box:-



$V=0$ for $0<x<L$

$V=\infty$ for $0>x>L$

$$\therefore \nabla^2\psi + \frac{2m}{\hbar^2}E\psi = 0$$

Let $\frac{2mE}{\hbar^2} = k^2$.

$$\Rightarrow \nabla^2\psi + k^2\psi = 0$$

Let the solution be $A\sin kx + B\cos kx = \psi$.

To find the value of A & B, we use the boundary condition.

So, at first boundary, at $x=0$, $\psi=0$.

$$\therefore A\sin kx + B\cos kx = \psi$$
$$\Rightarrow 0 + B\cos kx = 0$$
$$\Rightarrow B = 0.$$

At $x=L$, $\psi=0$

$$\Rightarrow A\sin kL + B\cos kL = 0$$
$$\Rightarrow A\sin kL = 0$$
$$\therefore \cancel{\sin} kL = n\pi \Rightarrow k = \frac{n\pi}{L}$$

$\frac{a^2}{a} - \frac{a^2}{a} - \frac{a^2}{4a}$

$x - \frac{a}{2}$

$$\boxed{\int \underbrace{\psi^*}_{\text{conjugate}} \psi = 1}$$

$\frac{x^3}{a^2} = \frac{a^3}{a^2} - \frac{a^3}{8a^2}$

$\frac{x^3}{3a^2}$ $\frac{a}{3} - \frac{a}{24}$ $\frac{7a}{8}$

Date : P. No. :

$$\therefore \quad k^2 = \frac{2mE}{\hbar^2}$$

$$\Rightarrow \quad \frac{n^2\pi^2}{L^2} = \frac{2mE}{\hbar^2}$$

$$\Rightarrow \quad E = \frac{n^2\pi^2\hbar^2}{2mL^2}$$

Now,

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{h/p} = \frac{p}{\hbar}$$

So, $p = \hbar k$

$$\Rightarrow p = \frac{n\pi\hbar}{L}$$

Now, $\psi = A\sin\left(\frac{n\pi x}{L}\right)$

& $\int_0^L \psi^* \psi = 1$

On solving, we get $A = \sqrt{\frac{2}{L}}$

$$\therefore \psi = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$$

Q. Find the probability in the region $\frac{a}{2} \le x \le a$ for a particle define by wave function,

$\psi(x) = \sqrt{\frac{3}{2a}}\left(1 - \frac{x}{a}\right)$

Sol^n → $P = \int_{a/2}^{a} \psi^2(x)\,dx = \int_{a/2}^{a} \frac{3}{2a}\left(1 - \frac{x}{a}\right)^2 dx$

$$\Rightarrow P = \frac{3}{2a}\int_{a/2}^{a}\left(1 - \frac{2x}{a} + \frac{x^2}{a^2}\right)dx = \frac{3}{2a}\left[\frac{a}{2} - \frac{3a}{4} + \frac{7a}{24}\right]$$

$$= \frac{3}{2}\left(\frac{1}{2} - \frac{3}{4} + \frac{7}{24}\right)$$

$$= \frac{3}{2}\left(\frac{12 - 18 + 7}{24}\right) = \frac{1}{16}$$

Date :
P. No. :

$\Rightarrow P = \frac{1}{16} = 0.0625$

Q. Compute the energy of the lowest 3 level for an electron in a square well of width 3Å. Given, $m_e = 9.1 \times 10^{-31}$ kg, n=1,2,3 & L = 3Å.

Q. The position and momentum of 0.5 keV are simultaneously determined. If its position is located with ~~the~~ in 0.2 nm, what is the percentage uncertainity in its momentum.

Sol^n → $\Delta p \times \Delta x \geq \frac{\hbar}{2}$

$\Rightarrow \Delta p \times 0.2 \times 10^{-9} \geq \frac{6.626 \times 10^{-34}}{4 \times 3.14}$

$\Rightarrow \Delta P = 2.63 \times 10^{-25}$ m

So, $\frac{\Delta P}{P} \times 100 = \frac{2.63 \times 10^{-25}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 0.5 \times 1000 \times 1.6 \times 10^{-19}}}$

$= 2.18\ \%$